

Observing the Critical Line in Split-Complex Coordinates

(Exercise XXVIII)

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Abstract

We observe the critical line of the Riemann direction through the split-complex frame (Exercise XXVII): the imaginary-radius orthogonal Euler circles. The observation yields three machine-checked results. First, the pseudo-norm of a critical-line point is $\|1/2 + tj\|^2 = 1/4 - t^2$: the real-part condition $\text{Re}(s) = 1/2$ becomes, in split coordinates, an imaginary-part threshold $|t| = 1/2$ — the two axes exchange roles. Second, the threshold is structural: points with $|t| < 1/2$ are timelike (inside the radius-1 circle) and points with $|t| > 1/2$ are spacelike (inside the radius- i circle); since the known zeros have $|t| \geq 14.13 > 1/2$, every known critical-line zero is spacelike in the split frame. Third, the reciprocal map sends the critical line to the hyperbola $(a-1)^2 - b^2 = 1$ — the split dual of the critical circle $(a-1)^2 + b^2 = 1$ already proved in the complex frame (C019–C022): the same map, two hosts, circle $\leftrightarrow$ hyperbola. Six theorems, zero errors, zero sorry, mathlib v4.32.2.

Keywords: Lean, formalization, split-complex, critical line, dual structure, 0pat

1 The observation

The split-complex frame (Exercise XXVII) hosts the imaginary-radius orthogonal Euler circles: $a+bj$, $j^2 = +1$, pseudo-norm $\|a+bj\|^2 = a^2 - b^2$. The radius-1 circle is the hyperbola $a^2 - b^2 = 1$; the radius- i circle is the conjugate hyperbola $a^2 - b^2 = -1$. We now observe the critical line of the Riemann direction — the line $\text{Re}(s) = 1/2$ — through this frame: its points are $\frac{1}{2} + tj$, $t \in \mathbb{R}$.

The observation is three theorems.

The pseudo-norm of the critical line. $\|\frac{1}{2} + tj\|^2 = \frac{1}{4} - t^2$. The real-part condition $\text{Re}(s) = 1/2$, which defines the critical line in the complex frame, becomes in the split frame an imaginary-part threshold: $|t| = 1/2$. The two axes exchange roles — the split metric swaps the meaning of the coordinates.

The threshold. Points with $t^2 < 1/4$ are timelike ($\|z\|^2 > 0$, inside the radius-1 circle); points with $t > 1/2$ are spacelike ($\|z\|^2 < 0$, inside the radius- i circle). Since the known zeros of ζ on the critical line have imaginary parts $|t| \geq 14.13 > 1/2$, every known critical-line zero is spacelike in the split frame — the radius- i region is where the zeros live. The threshold $\frac{1}{2}$ is not chosen; it is the value of the frame.

The dual of the critical circle. The reciprocal map $z \mapsto z^{-1}$ (the split inverse, valid when $\|z\|^2 \neq 0$) sends the critical line to the hyperbola $(a-1)^2 - b^2 = 1$. In the complex frame, the same reciprocal map sends the critical line to the circle $(a-1)^2 + b^2 = 1$ — the critical circle, proved in the Riemann chain (C019–C022). The split observation is the exact dual: the same map, two hosts, circle $\leftrightarrow$ hyperbola. The center $(1, 0)$ and the radius 1 are unchanged; only the metric changes.

2 Honest boundary and position

The content is classical mathematics (split-complex algebra, elementary computations); the value is observational: the critical-line geometry of the Riemann direction has a dual structure in the split frame, and the dual places the zeros (if on the critical line) in the spacelike region with a structural threshold $\frac{1}{2}$. No claim is made about the zeros themselves: the observation concerns the geometry of the line, not the analysis of ζ . The frame observes; the analysis remains open.

3 Formalization

The proof is a single Lean file (proof.lean, 127 lines), importing only Mathlib.Data.Real.Basic and basic tactics. The split-complex structure (30 lines, as in Exercise XXVII) is self-contained. Six theorems, compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry.

Theorem	Statement
inv_mul_self	$z^{-1}z = 1$ ($\ z\ ^2 \neq 0$)
critical_line_pseudonorm	$\ \frac{1}{2} + tj\ ^2 = \frac{1}{4} - t^2$
threshold_timelike	$t^2 < 1/4 \Rightarrow \ z\ ^2 > 0$
threshold_spacelike	$t > 1/2 \Rightarrow \ z\ ^2 < 0$
zero_observation_spacelike	$ t > 1/2 \Rightarrow \ z\ ^2 < 0$
recip_critical_line_hyperbola	$\text{recip}(\frac{1}{2} + tj)$ on $(a-1)^2 - b^2 = 1$

4 Conclusion

The critical line, observed through the imaginary-radius Euler circles, reveals its dual: the circle of the complex frame becomes the hyperbola of the split frame under the same reciprocal map, and the threshold $\frac{1}{2}$ appears as a structural value of the metric. The zeros of ζ (if on the critical line) live in the spacelike region, inside the radius- i circle. The observation is complete; the analysis that would place the zeros there remains the open span of the bridge.

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